

Object Oriented Programming (CSC166)

Lab Sheet 3

1. Write a program that defines a class "Counter" with a private data member count. Overload the ++ operator (both prefix and postfix forms) as member functions to increment count by 1, and write a member function to display the current value of count.
*Hint: [Counter(int c = 0) → constructor that initializes count
Counter operator ++ () → prefix version, increments count then returns the object
Counter operator ++ (int) → postfix version, returns the old value then increments count
displayCount() → prints the current value of count]*
2. Write a program that defines a class "Complex" with private data members real and imag. Overload the binary + operator as a member function to add two Complex objects.
*Hint: [Complex() → default constructor, initializes real and imag to 0
Complex(float r, float i) → parameterized constructor
Complex operator + (Complex c2) → member function that adds the real parts and imaginary parts of two Complex objects
displayComplex() → prints the complex number in the form a + bi]*
3. Write a program that defines a class "Distance" with private data members feet and inches. Overload the binary + operator using friend function to add two Distance objects. Handle the case where the resulting inches value is 12 or more by carrying over into feet.
*Hint: [Distance() → default constructor, initializes feet and inches to 0
Distance(int f, int i) → parameterized constructor
Distance operator + (Distance d2) → friend function that adds feet and inches, carries over 12 inches into 1 foot
displayDistance() → prints the distance in the form feet'-inches"]*
4. Write a program that defines a class "Time" with private data members hours and minutes. Overload the binary + operator as a friend function to add two Time objects, and overload the == operator as a friend function to check whether two Time objects are equal.
*Hint: [friend Time operator + (Time t1, Time t2) → adds hours and minutes, carries over 60 minutes into 1 hour
friend bool operator == (Time t1, Time t2) → returns true if hours and minutes of both objects match
displayTime() → prints the time in the form hh:mm]*